

Practical week-1

Query 1: Create the database

The screenshot shows the SQL Developer interface. On the left, the 'SCHEMAS' pane lists 'joins', 'klu', 'pweek2', and 'sys'. The main editor window contains a single SQL statement: 'create database pweek1'. The 'Output' pane at the bottom displays the execution results in a table format.

#	Time	Action	Message	Duration / Fetch
1	15:11:22	create database pweek1	1 row(s) affected	0.016 sec

Query 2: Select the database

The screenshot shows the SQL Developer interface with two SQL statements in the editor: 'create database pweek1' and 'use pweek1'. The 'Output' pane at the bottom displays the execution results for both statements.

#	Time	Action	Message	Duration / Fetch
1	15:11:22	create database pweek1	1 row(s) affected	0.016 sec
2	15:13:25	use pweek1	0 row(s) affected	0.000 sec

Query 3: Create the books table

Navigator: SCHEMAS
Filter objects
joins
klu
pweek2
sys

Administration Schemas
Information

No object selected

SQL File 3* SQL File 3* SQL File 4* x

Limit to 1000 rows

SQLAdditions: My Snippets

```

1 create database pweek1
2 use pweek1
3
4 CREATE TABLE books (
5     book_id
6     INT AUTO_INCREMENT PRIMARY KEY,
7     title
8     VARCHAR(255) NOT NULL,
9     isbn
10    VARCHAR(13) NOT NULL UNIQUE,
11    published_year INT,
12    CONSTRAINT chk_published_year CHECK (published_year < 2027)
13 );

```

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 1	15:11:22	create database pweek1	1 row(s) affected	0.016 sec
✓ 2	15:13:25	use pweek1	0 row(s) affected	0.000 sec
✓ 3	15:14:08	CREATE TABLE books (book_id INT AUTO_INC...	0 row(s) affected	0.109 sec

Query 4: Create the members table

Navigator: SCHEMAS
Filter objects
joins
klu
pweek2
sys

Administration Schemas
Information

No object selected

SQL File 3* SQL File 3* SQL File 4* x

Limit to 1000 rows

SQLAdditions: My Snippets

```

3
4 CREATE TABLE books (
5     book_id
6     INT AUTO_INCREMENT PRIMARY KEY,
7     title
8     VARCHAR(255) NOT NULL,
9     isbn
10    VARCHAR(13) NOT NULL UNIQUE,
11    published_year INT,
12    CONSTRAINT chk_published_year CHECK (published_year < 2027)
13 );
14
15 CREATE TABLE members (
16     member_id INT AUTO_INCREMENT PRIMARY KEY,
17     full_name VARCHAR(100) NOT NULL,
18     email
19     VARCHAR(150) NOT NULL UNIQUE
20 );

```

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 1	15:11:22	create database pweek1	1 row(s) affected	0.016 sec
✓ 2	15:13:25	use pweek1	0 row(s) affected	0.000 sec
✓ 3	15:14:08	CREATE TABLE books (book_id INT AUTO_INC...	0 row(s) affected	0.109 sec
✓ 4	15:14:56	CREATE TABLE members (member_id INT AUTO_INC...	0 row(s) affected	0.079 sec

Query 5 (verification): Inspect the table structures

Navigator: SCHEMAS
Filter objects
joins
klu
pweek2
sys

SQL File 3* SQL File 3* SQL File 4* x
Limit to 1000 rows

```

13 );
14
15 CREATE TABLE members (
16     member_id INT AUTO_INCREMENT PRIMARY KEY,
17     full_name VARCHAR(100) NOT NULL,
18     email
19     VARCHAR(150) NOT NULL UNIQUE
20 );
21
22 describe books;

```

Result Grid

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	<u>NONE</u>	auto_increment
title	varchar(255)	NO		<u>NONE</u>	
isbn	varchar(13)	NO	UNI	<u>NONE</u>	
published_year	int	YES		<u>NONE</u>	

Administration Schemas
Information

No object selected

Result 1 x Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	15:11:22	create database pweek1	1 row(s) affected	0.016 sec
2	15:13:25	use pweek1	0 row(s) affected	0.000 sec
3	15:14:08	CREATE TABLE books (book_id INT AUTO_INC...	0 row(s) affected	0.109 sec
4	15:14:56	CREATE TABLE members (member_id INT AUTO_IN...	0 row(s) affected	0.079 sec
5	15:18:47	describe books	4 row(s) returned	0.016 sec / 0.000 sec

Query 6: Insert 3 books

Navigator: SCHEMAS
Filter objects
joins
klu
pweek2
sys

SQL File 3* SQL File 3* SQL File 4* x
Limit to 1000 rows

```

10 VARCHAR(13) NOT NULL UNIQUE,
11 published_year INT,
12 CONSTRAINT chk_published_year CHECK (published_year < 2027)
13 );
14
15 CREATE TABLE members (
16     member_id INT AUTO_INCREMENT PRIMARY KEY,
17     full_name VARCHAR(100) NOT NULL,
18     email
19     VARCHAR(150) NOT NULL UNIQUE
20 );
21
22 describe books;
23
24 INSERT INTO books (title, isbn, published_year) VALUES
25 ('The Alchemist', '9780061122415', 1988),
26 ('Clean Code', '9780132350884', 2008),
27 ('Atomic Habits', '9780735211292', 2018);

```

Administration Schemas
Information

No object selected

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	15:11:22	create database pweek1	1 row(s) affected	0.016 sec
2	15:13:25	use pweek1	0 row(s) affected	0.000 sec
3	15:14:08	CREATE TABLE books (book_id INT AUTO_INC...	0 row(s) affected	0.109 sec
4	15:14:56	CREATE TABLE members (member_id INT AUTO_IN...	0 row(s) affected	0.079 sec
5	15:18:47	describe books	4 row(s) returned	0.016 sec / 0.000 sec
6	15:20:03	INSERT INTO books (title, isbn, published_year) VALU...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec

Query 7: Insert 3 members

Navigator

SQL File 3* SQL File 3* SQL File 4*

Limit to 1000 rows

SQLAdditions

My Snippets

SCHEMAS

Filter objects

joins

klu

pweek2

sys

Administration Schemas

Information

No object selected

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	15:11:22	create database pweek1	1 row(s) affected	0.016 sec
2	15:13:25	use pweek1	0 row(s) affected	0.000 sec
3	15:14:08	CREATE TABLE books (book_id INT AUTO_INC...	0 row(s) affected	0.109 sec
4	15:14:56	CREATE TABLE members (member_id INT AUTO_IN...	0 row(s) affected	0.079 sec
5	15:18:47	describe books	4 row(s) returned	0.016 sec / 0.000 sec
6	15:20:03	INSERT INTO books (title, isbn, published_year) VALU...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
7	15:20:44	INSERT INTO members (full_name, email) VALUES (A...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.031 sec

Object Info Session

Test 1: Duplicate ISBN (UNIQUE violation)

Navigator

SCHEMAS

Filter objects

- joins
- klu
- pweek2
- sys

Administration Schemas

Information

No object selected

SQL File 3* SQL File 3* SQL File 4* x

Limit to 1000 rows

SQLAdditions

My Snippets

```

18 email
19 VARCHAR(150) NOT NULL UNIQUE
20 );
21
22 • describe books;
23
24 • INSERT INTO books (title, isbn, published_year) VALUES
25 ('The Alchemist', '9780061122415', 1988),
26 ('Clean Code', '9780132350884', 2008),
27 ('Atomic Habits', '9780735211292', 2018);
28
29 • INSERT INTO members (full_name, email) VALUES
30 ('Anil Kumar', 'anil.kumar@example.com'),
31 ('Priya Sharma', 'priya.sharma@example.com'),
32 ('Ravi Verma', 'ravi.verma@example.com');
33
34 • INSERT INTO books (title, isbn, published_year)
35 VALUES ('Fake Copy', '9780061122415', 2000);

```

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
2	15:13:25	use pweek1	0 row(s) affected	0.000 sec
3	15:14:08	CREATE TABLE books (book_id INT AUTO_I...	0 row(s) affected	0.109 sec
4	15:14:56	CREATE TABLE members (member_id INT AUTO_I...	0 row(s) affected	0.079 sec
5	15:18:47	describe books	4 row(s) returned	0.016 sec / 0.000 sec
6	15:20:03	INSERT INTO books (title, isbn, published_year) VAL...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
7	15:20:44	INSERT INTO members (full_name, email) VALUES (...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.031 sec
8	15:23:22	INSERT INTO books (title, isbn, published_year) VAL...	Error Code: 1062. Duplicate entry '9780061122415' fo...	0.000 sec

Object Info Session

Test 2: NULL title (NOT NULL violation)

Navigator

SCHEMAS

Filter objects

- joins
- klu
- pweek1
- pweek2
- sys

Administration Schemas

Information

No object selected

SQL File 3* SQL File 3* SQL File 4* x

Limit to 1000 rows

SQLAdditions

My Snippets

```

21
22 • describe books;
23
24 • INSERT INTO books (title, isbn, published_year) VALUES
25 ('The Alchemist', '9780061122415', 1988),
26 ('Clean Code', '9780132350884', 2008),
27 ('Atomic Habits', '9780735211292', 2018);
28
29 • INSERT INTO members (full_name, email) VALUES
30 ('Anil Kumar', 'anil.kumar@example.com'),
31 ('Priya Sharma', 'priya.sharma@example.com'),
32 ('Ravi Verma', 'ravi.verma@example.com');
33
34 • INSERT INTO books (title, isbn, published_year)
35 VALUES ('Fake Copy', '9780061122415', 2000);
36
37 • INSERT INTO books (title, isbn, published_year)
38 VALUES (NULL, '9999999999999', 2010);

```

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
3	15:14:08	CREATE TABLE books (book_id INT AUTO_I...	0 row(s) affected	0.109 sec
4	15:14:56	CREATE TABLE members (member_id INT AUTO_I...	0 row(s) affected	0.079 sec
5	15:18:47	describe books	4 row(s) returned	0.016 sec / 0.000 sec
6	15:20:03	INSERT INTO books (title, isbn, published_year) VAL...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
7	15:20:44	INSERT INTO members (full_name, email) VALUES (...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.031 sec
8	15:23:22	INSERT INTO books (title, isbn, published_year) VAL...	Error Code: 1062. Duplicate entry '9780061122415' fo...	0.000 sec
9	15:24:36	INSERT INTO books (title, isbn, published_year) VAL...	Error Code: 1048. Column 'title' cannot be null	0.000 sec

Object Info Session

Test 3: Future publication year (CHECK violation)

Navigator: SCHEMAS

Filter objects

- joins
- klu
- pweek1
- pweek2
- sys

Administration Schemas

Information

No object selected

SQL File 3* SQL File 3* SQL File 4* x

Limit to 1000 rows

My Snippets

```

24 • INSERT INTO books (title, isbn, published_year) VALUES
25 ('The Alchemist', '9780061122415', 1988),
26 ('Clean Code', '9780132350884', 2008),
27 ('Atomic Habits', '9780735211292', 2018);
28
29 • INSERT INTO members (full_name, email) VALUES
30 ('Anil Kumar', 'anil.kumar@example.com'),
31 ('Priya Sharma', 'priya.sharma@example.com'),
32 ('Ravi Verma', 'ravi.verma@example.com');
33
34 • INSERT INTO books (title, isbn, published_year)
35 VALUES ('Fake Copy', '9780061122415', 2000);
36
37 • INSERT INTO books (title, isbn, published_year)
38 VALUES (NULL, '9999999999999', 2010);
39
40 • INSERT INTO books (title, isbn, published_year)
41 VALUES ('Time Traveler', '8888888888888', 2030);

```

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
4	15:14:56	CREATE TABLE members (member_id INT AUTO_I...	0 row(s) affected	0.079 sec
5	15:18:47	describe books	4 row(s) returned	0.016 sec / 0.000 sec
6	15:20:03	INSERT INTO books (title, isbn, published_year) VAL...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
7	15:20:44	INSERT INTO members (full_name, email) VALUES (...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.031 sec
8	15:23:22	INSERT INTO books (title, isbn, published_year) VAL...	Error Code: 1062. Duplicate entry '9780061122415' fo...	0.000 sec
9	15:24:36	INSERT INTO books (title, isbn, published_year) VAL...	Error Code: 1048. Column 'title' cannot be null	0.000 sec
10	15:25:18	INSERT INTO books (title, isbn, published_year) VAL...	Error Code: 3819. Check constraint 'chk_published_y...	0.000 sec

Object Info Session

Test 4: Duplicate email (UNIQUE violation)

Navigator: SCHEMAS

Filter objects

- joins
- klu
- pweek1
- pweek2
- sys

Administration Schemas

Information

No object selected

SQL File 3* SQL File 3* SQL File 4* x

Limit to 1000 rows

My Snippets

```

27 ('Atomic Habits', '9780735211292', 2018);
28
29 • INSERT INTO members (full_name, email) VALUES
30 ('Anil Kumar', 'anil.kumar@example.com'),
31 ('Priya Sharma', 'priya.sharma@example.com'),
32 ('Ravi Verma', 'ravi.verma@example.com');
33
34 • INSERT INTO books (title, isbn, published_year)
35 VALUES ('Fake Copy', '9780061122415', 2000);
36
37 • INSERT INTO books (title, isbn, published_year)
38 VALUES (NULL, '9999999999999', 2010);
39
40 • INSERT INTO books (title, isbn, published_year)
41 VALUES ('Time Traveler', '8888888888888', 2030);
42
43 • INSERT INTO members (full_name, email)
44 VALUES ('Anil Clone', 'anil.kumar@example.com');

```

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
5	15:18:47	describe books	4 row(s) returned	0.016 sec / 0.000 sec
6	15:20:03	INSERT INTO books (title, isbn, published_year) VAL...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
7	15:20:44	INSERT INTO members (full_name, email) VALUES (...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.031 sec
8	15:23:22	INSERT INTO books (title, isbn, published_year) VAL...	Error Code: 1062. Duplicate entry '9780061122415' fo...	0.000 sec
9	15:24:36	INSERT INTO books (title, isbn, published_year) VAL...	Error Code: 1048. Column 'title' cannot be null	0.000 sec
10	15:25:18	INSERT INTO books (title, isbn, published_year) VAL...	Error Code: 3819. Check constraint 'chk_published_y...	0.000 sec
11	15:26:06	INSERT INTO members (full_name, email) VALUES (...	Error Code: 1062. Duplicate entry 'anil.kumar@examp...	0.032 sec